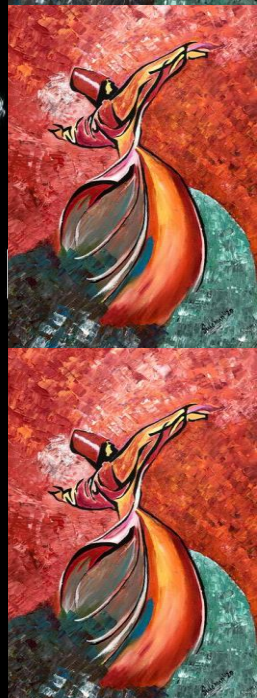




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



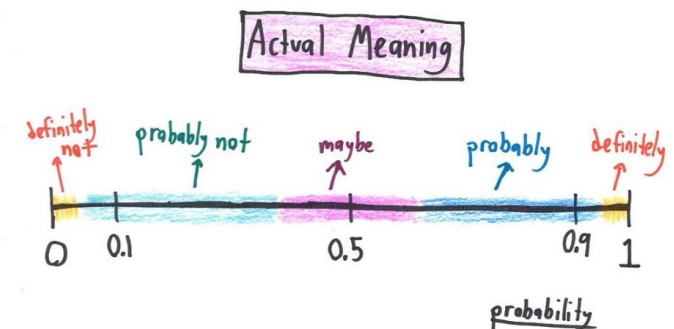


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 7 - Lecture No 13 – 1st October 2025



Announcements!

1. Lectures up to No 11 posted
2. To enable N-2. we will like to have a quiz No 4 next week?
3. Midterm Exam - October 8, 2025, from 05:40 PM to 06:55 PM – **venue H-507 (L2)** – Good Luck

Agenda for today

1. Unit 4 - Discrete Random Variables

Unit 4
Discrete Random Variables

Variance of Bernoulli r.v

A useful formula: $\text{var}(X) = E[X^2] - (E[X])^2$

$$X = \begin{cases} 1, & \text{w. p. } p \\ 0, & \text{w. p. } 1 - p \end{cases}$$

$$E[X] = 1(p) + 0(1 - p) = p$$

$$X^2 = \begin{cases} 1, & \text{w. p. } p \\ 0, & \text{w. p. } 1 - p \end{cases}$$

$$E[X^2] = 1(p) + 0(1 - p) = p$$

$$\begin{aligned} \text{var}(X) &= E[X^2] - (E[X])^2 \\ &= p - p^2 \\ &= p(1 - p) \end{aligned}$$

Exercise –

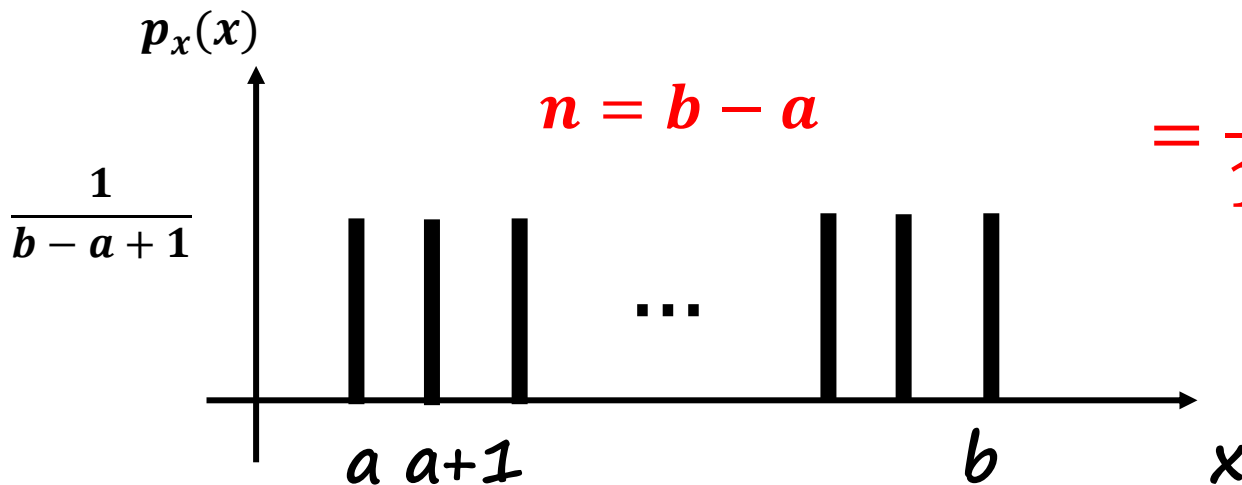
- i. Plot $\text{Var}(X)$ for different possible values of p .
- ii. What does the plot tell you?
- iii. What does the plot tell you about the fair coin?

Variance of the uniform



$$\text{var}(X) = E[X^2] - (E[X])^2 = \frac{1}{n+1} \left(0^2 + 1^2 + 2^2 + \dots + n^2 \right) - \left(\frac{n}{2} \right)^2$$

$$\frac{1}{6}n(n+1)(2n+1)$$



$$n = b - a$$

$$= \frac{1}{12}n(n+2)$$

$$\text{Var}(x) = \frac{1}{12}(b-a)(b-a+2)$$

Conditional PMF and expectation, given an event

- *Condition on an event $A \rightarrow$ use conditional probabilities*

$$p_X(x) = P(X = x)$$

$$\underline{p_{X|A}(x)} = \underline{P(X = x|A)} \quad \text{assume } P(A) > 0$$

$$\sum_x p_X(x) = 1$$

$$\sum_x p_{X|A}(x) = 1$$

$$E[X] = \sum_x x p_X(x)$$

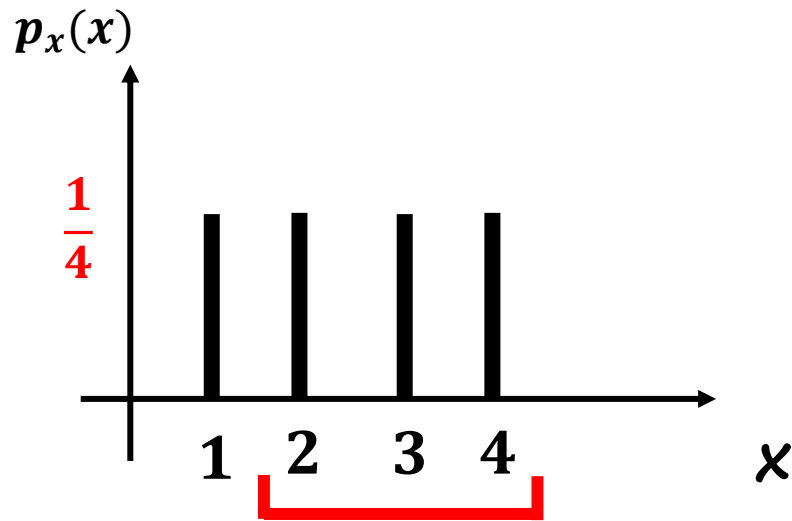
$$E[X|A] = \quad ?$$

$$E[g(X)] = \sum_x g(x) p_X(x)$$

$$E[g(X)|A] = \sum_x g(x) p_{X|A}(x)$$

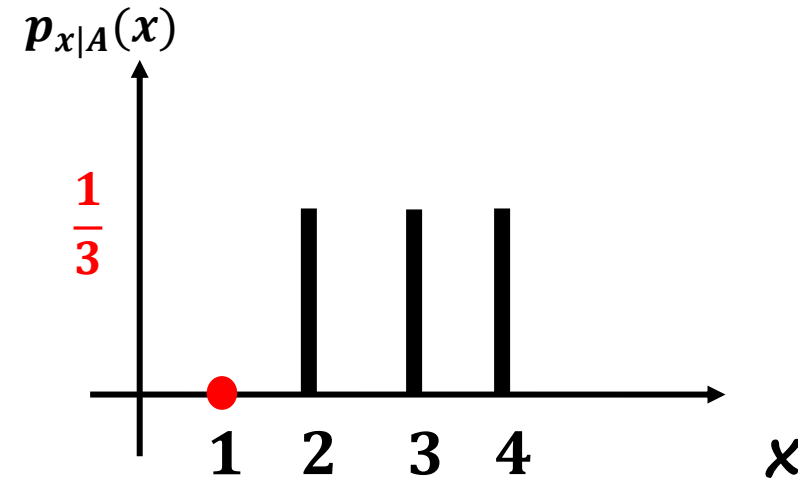
Example of conditioning

- Let $A = \{X \geq 2\}$



$$E[X] = 2.5$$

$$\begin{aligned}\text{var}[X] &= \frac{1}{12}(b-a)(b-a+2) \\ &= \frac{1}{12} * 3 * 5 = \frac{5}{4}\end{aligned}$$

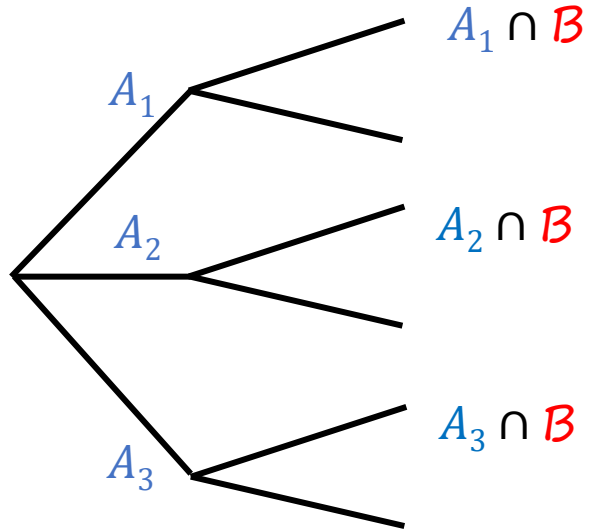


$$E[X|A] = 3$$

$$\text{var}[X|A] =$$

$$\frac{1}{3}(4-3)^2 + \frac{1}{3}(3-3)^2 + \frac{1}{3}(2-3)^2 = \frac{2}{3}$$

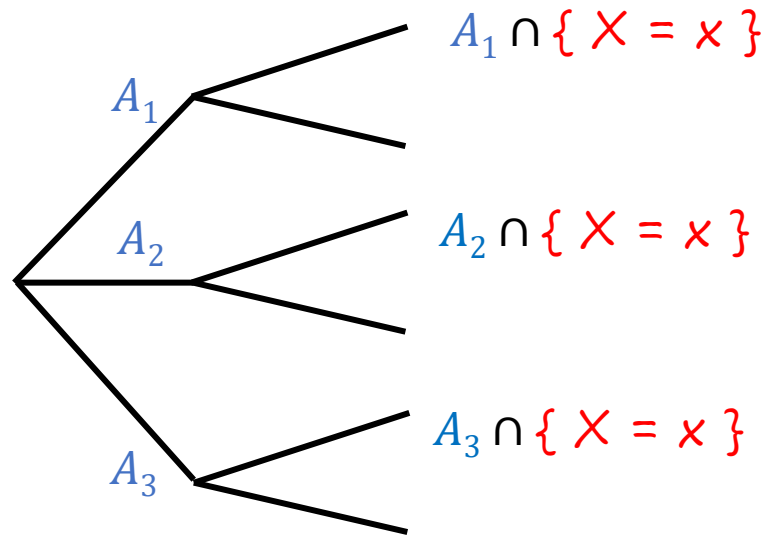
Total Expectation Theorem



$$\mathbf{P}(B) = \mathbf{P}(A_1) \mathbf{P}(B | A_1) + \dots + \mathbf{P}(A_n) \mathbf{P}(B | A_n)$$

$$B = \{X = x\}$$

Total Expectation Theorem

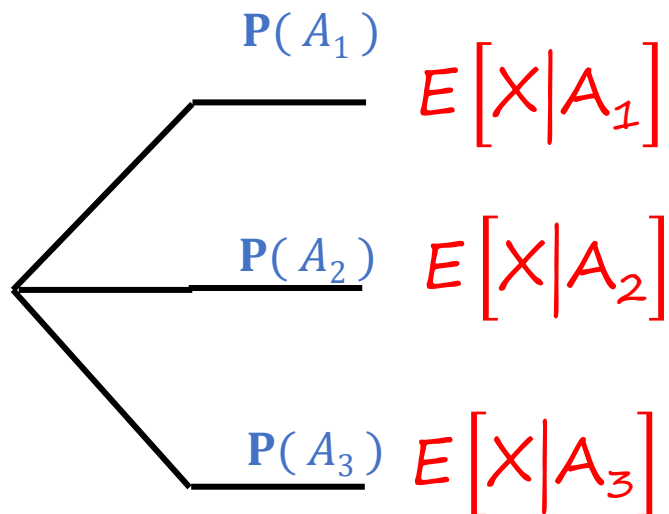


$$\mathbf{P}(B) = \mathbf{P}(A_1) \mathbf{P}(B | A_1) + \dots + \mathbf{P}(A_n) \mathbf{P}(B | A_n)$$

$$B = \{X = x\}$$

$$p_X(x) = \mathbf{P}(A_1) p_{X|A_1}(x) + \dots + \mathbf{P}(A_n) p_{X|A_n}(x)$$

for all x



$$= \underbrace{\sum_x x p_X(x)}_{E[X]} = \mathbf{P}(A_1) \underbrace{\sum_x x p_{X|A_1}(x)}_{E[X|A_1]} + \dots$$

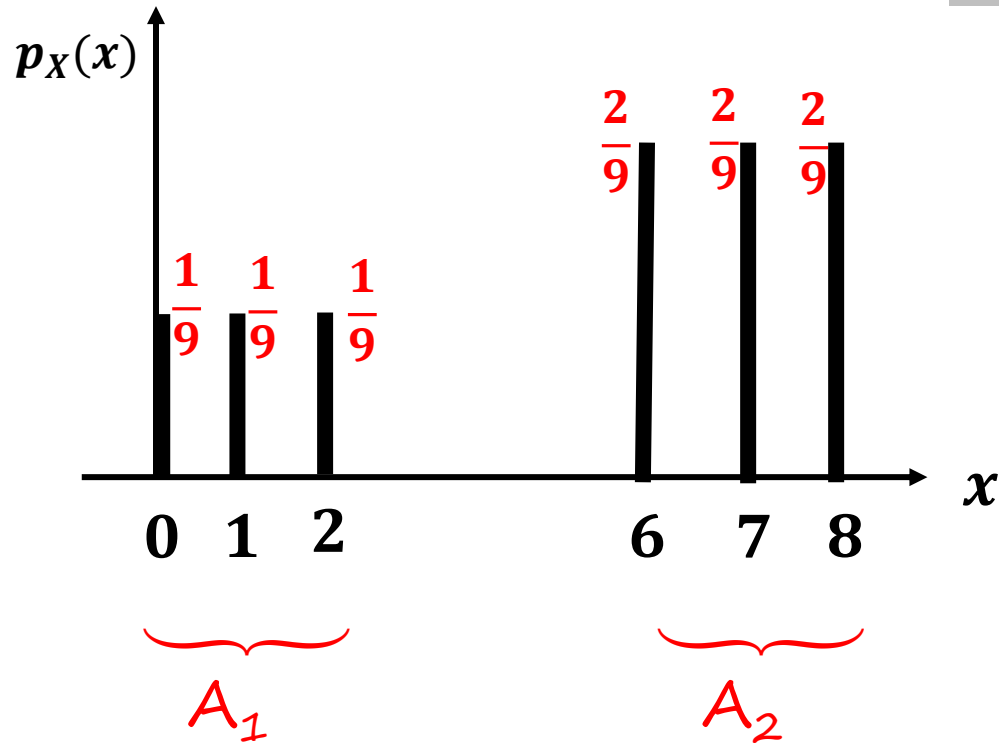
$$E[X] = \mathbf{P}(A_1) E[X | A_1] + \dots + \mathbf{P}(A_n) E[X | A_n]$$

Total Expectation Theorem

Total Expectation Example

Exercise No 01!

Using Total expectation theorem, Calculate the $E[X]$



$$\Rightarrow P(A_1) = 1/3$$

$$\Rightarrow P(A_2) = 2/3$$

$$\Rightarrow E[X | A_1] = 1$$

$$\Rightarrow E[X | A_2] = 7$$

$$\Rightarrow E[X] = \frac{1}{3} * 1 + \frac{2}{3} * 7$$

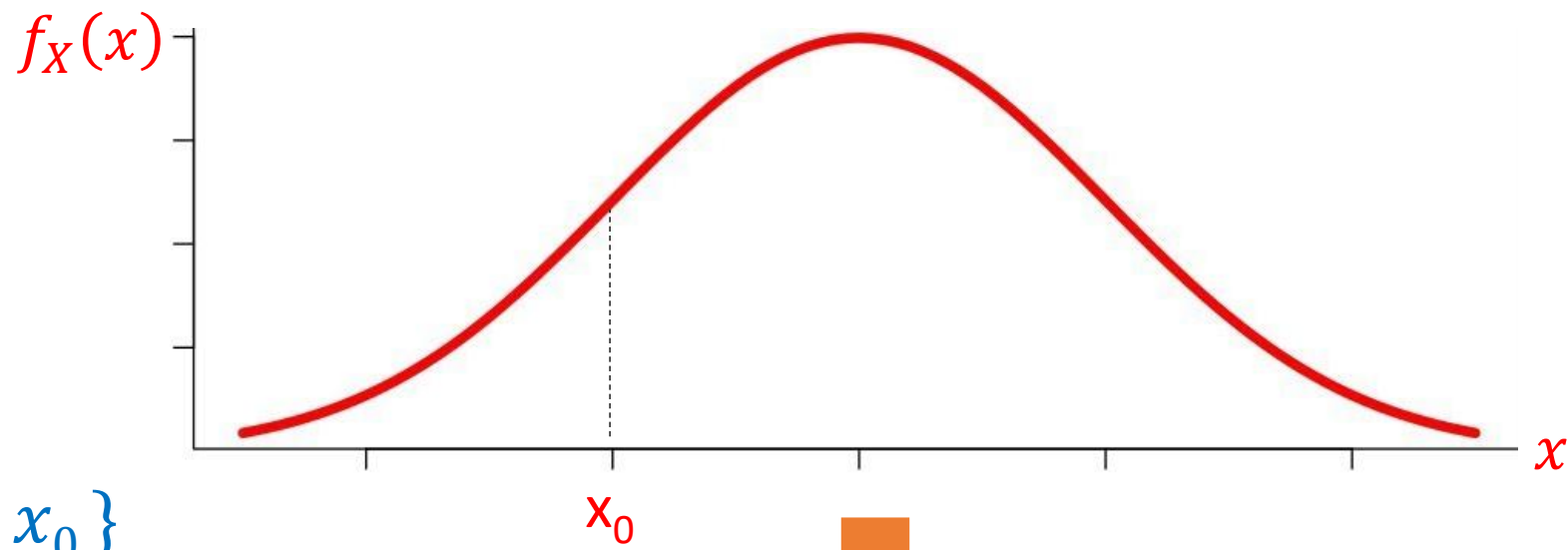
$$E[X] = P(A_1) E[X | A_1] + \dots + P(A_n) E[X | A_n]$$

Total Expectation Theorem

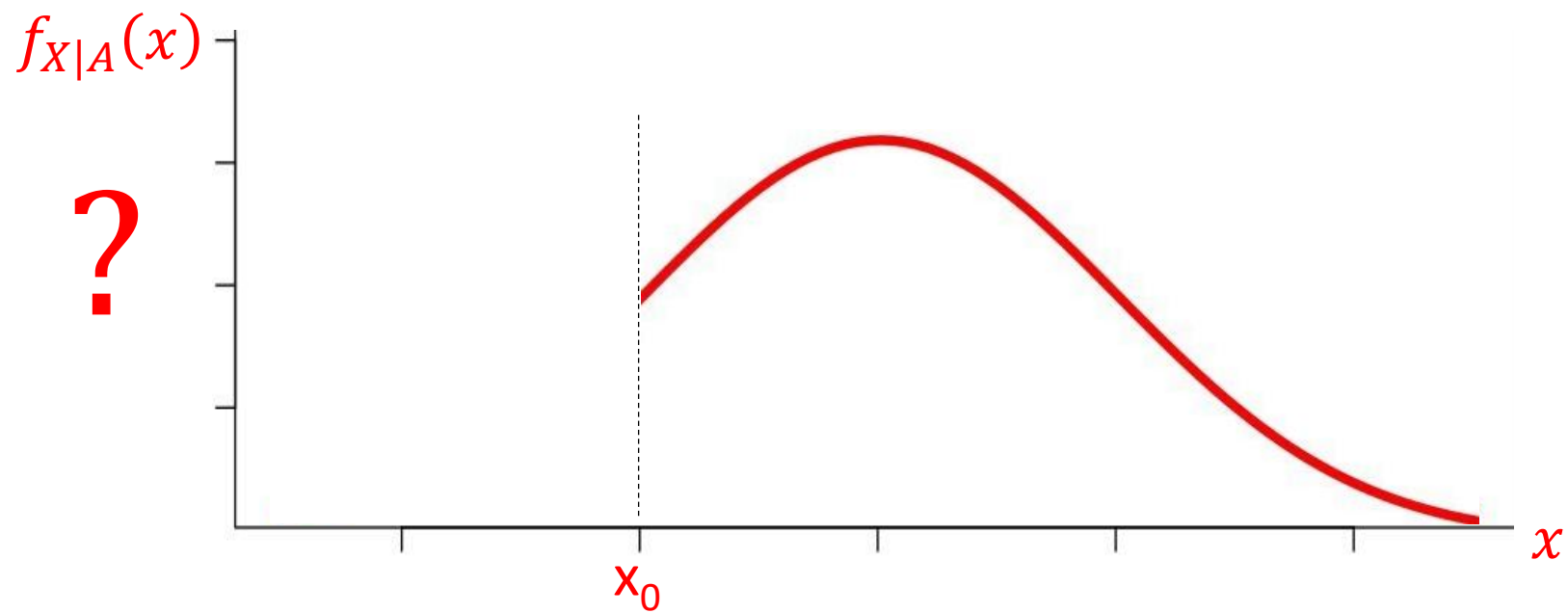
Re-visiting the great question by a former student!



“Does the nature of distribution after conditioning remains the same?”

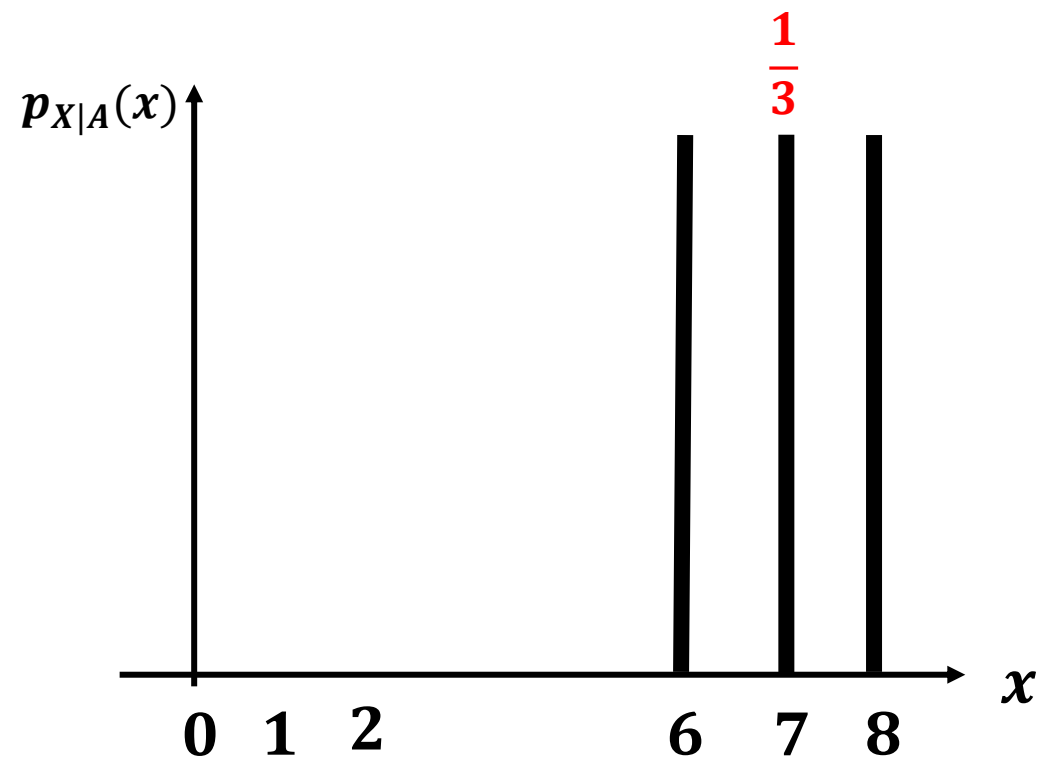
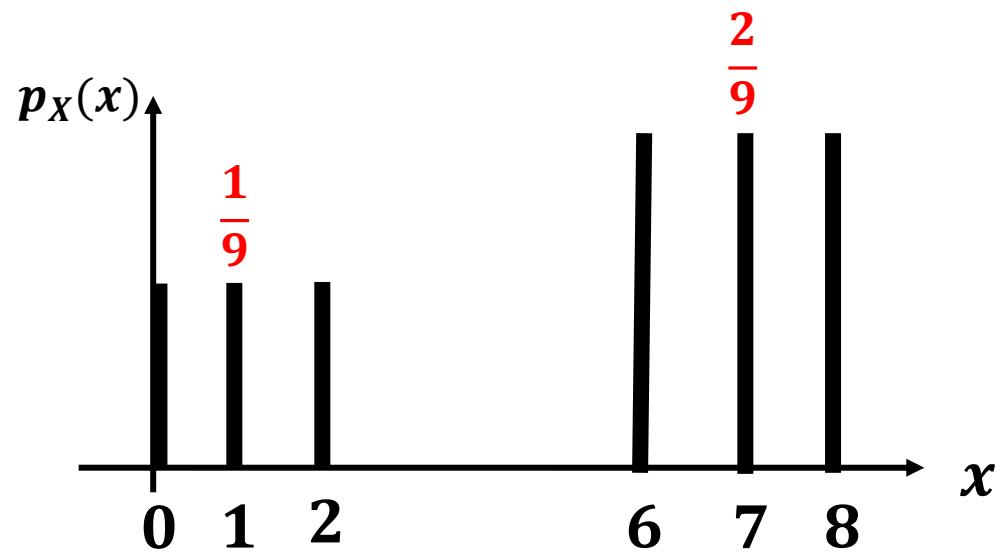


$$A = \{x > x_0\}$$



$$A = \{x > 2\}$$

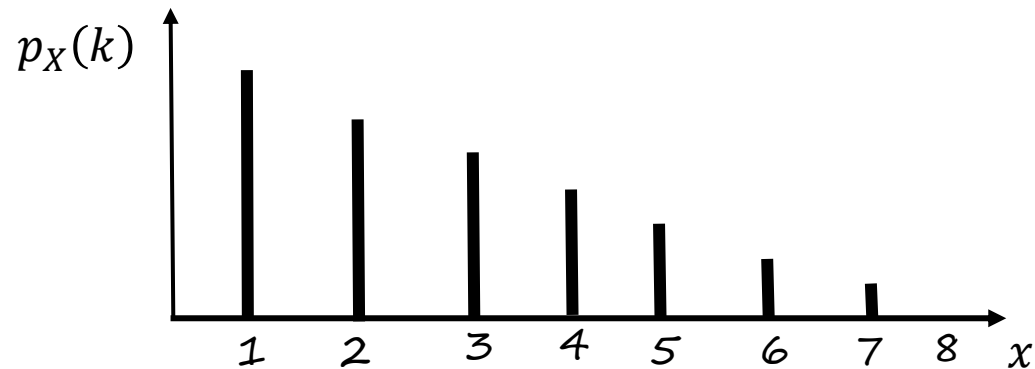
$$P_{X|A}(x) \text{ ?}$$



Conditioning a geometric R.V.

- **X**: number independent tosses of a coin until first heads with $P(\text{Heads}) = p$

$$p_X(x) = (1 - p)^{x-1}p \quad x = 1, 2, 3, \dots$$



Memorylessness:

Number of **remaining** coin tosses, conditioned on Tails in the first toss, is **Geometric**, with parameter p

Conditioned on $X > 1$, $X - 1$ is geometric with parameter p

